

TrES-1b

R-2019

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_121001 · created 2026-08-07T23:55:01+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456201.72748 +/- 0.0009 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1751 +/- 0.0073
Transit depth (Rp/Rs)^2	3.07 +/- 0.25 [%]
Orbital Inclination (inc)	89.07 +/- 0.65
Airmass coefficient 1 (a1)	1.326 +/- 0.021
Airmass coefficient 2 (a2)	-0.1369 +/- 0.0029
Scatter in the residuals of the lightcurve fit is	1.54 %
Best Comparison Star	None
Optimal Aperture	3.58
Optimal Annulus	9.33
Transit Duration (day)	0.1063 +/- 0.0019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1751 $\pm$ 0.0073 vs 0.1358 (deviation 3.85 σ)
Duration fit vs published	0.1063 d vs 0.1042 d (deviation 0.8 σ)
Mid-time O-C	0.8 min
Depth SNR	17.2
Residual scatter	1.54 %

Light curve

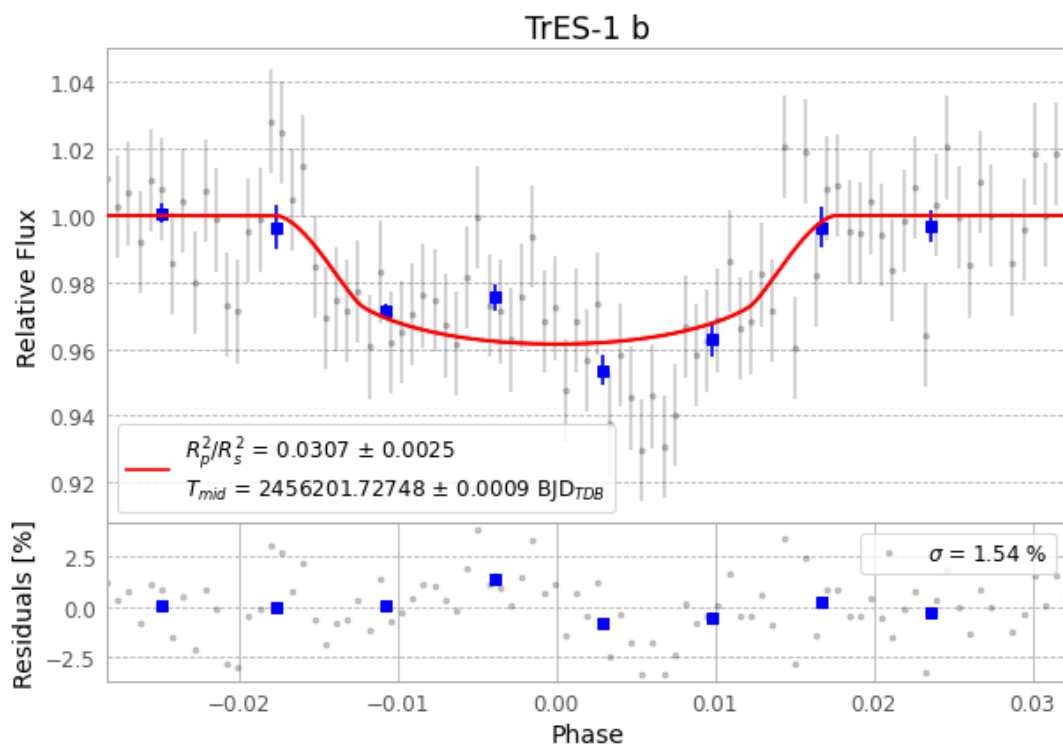


Figure 1 — FinalLightCurve_TrES-1 b_2012-09-30.png (SHA-256 b52590207a5d0079...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [268, 263], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 56fbb96a099cdc17...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

89 FITS frames (58 MB), TRES-1121001032112.FITS → TRES-1121001074512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-121001.log (d90032b4b195...), FinalLightCurve_TrES-1 b_2012-09-30.png (b52590207a5d...), FinalLightCurve_TrES-1 b_2012-09-30.csv (93481cb7024e...)

Compute resources

Wall time 157.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 23:55Z.